Example Data Analysis Final Report

Samuel Haws

Abstract—This report analyzes a mock e-commerce dataset covering 5000 users and 9381 site decisions, examining traffic sources, conversion ratios, and product performance to draw financial, marketing, and website-design conclusions.

1. INTRODUCTION

Data analysis is an important skill in understanding and using data, this report is part of a full repository built from a mock dataset to improve both Python-based analytical skills and reporting skills. In this investigation, the dataset is a .csv spreadsheet from an online business which drives its customers from multiple different streams. We are also given customer movements throughout the site which proves useful for understanding customer behaviour.

Files within this repository:

- `website_analysis.ipynb` - Jupyter Notebook workflow, with light explanation and output charts.
- `site_data.csv` - Data collected from the site that is used for the analysis.
- `website_analysis_report.tex` - This current report. An in detail report of the analysis.
- `website_analysis_take.tex` — Simpler presentation involving key takeaways.

This investigation was influenced by research that began with `roadmap.io` and led into IBM's course on data analytics.

1.1. Python Libraries

The libraries used are as follows: `matplotlib` `math` `pandas`. They are all imported first for better presentation of code.

2. DATA PREPARATION

2.1. Importing Data

The benefit of using python as a tool for analysis is it directly reads the data file, so less risk of formatting problems. The panda library was used to read and head the first lines to check for any importing faults in which none were found.

2.2. Cleaning

To reduce work flaws, cleaning data is essential, otherwise results could be skewed and incorrect which could make any conclusion harmful. Fortunately, there are two easy steps we used for verifying clean data:
1. Duplicate Rows - which returned as false.
2. Empty Rows - which also returned as false.
Since both came back as false we can conclude there is no missing or false data.

3. DESCRIPTIVE ANALYSIS

With a clean sheet we can now begin to understand our data. Basic maths tools are used to find the scale of the data we are working with, so that the conclusions made can be applied appropriately to the business.

3.1. User Orders

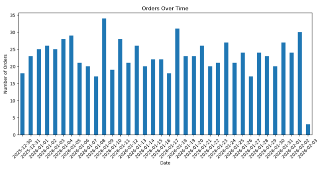
Traffic drives a total user count of 5000 making a total amount of 9381 decisions within the given time period. Of these 9381 decisions, 826 are final purchases, bringing in a total revenue of 87975.11 for the company.

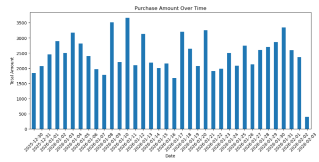
Basic maths then tells us that the average amount spent per user is 106.51, and through further investigation that the highest and lowest orders are 199.61 and 20.18 respectively.

3.2. Time Orders

A company's progress can be modeled well by seeing how its stats change over time. We may have calculated the total orders, but it would be better understanding how it changes over time. By graphing it we can assess visually if there are any trends or cycles that affect consumer decisions. As shown, orders remain at a pretty consistent level on a daily basis.

We can do the same with purchase amount, although it appears more cyclical, we arrive at the same conclusion.

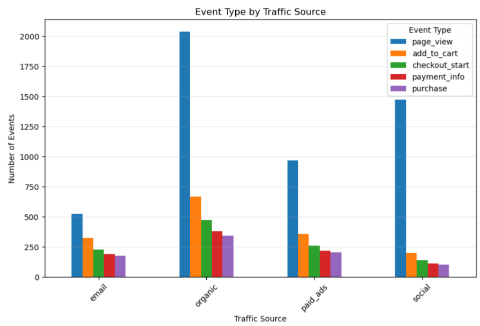




4. TRAFFIC SOURCE ANALYSIS

4.1. Traffic Choices

The initial workflow for this section led us to bar charts showing views and then purchases by traffic source. We later realised that a stacked bar chart would let us understand it in more detail.



We can see that the most traffic is brought in by organic and social media means. However their ratio of bringing in revenue is a lot less. We can already begin to make assumptions that customers find out about the site through other means due to such a high organic view count, the next highest being social media, which makes sense given typical scrolling behaviour. Paid ads appear to be somewhat effective, but would only be profitable if the amount spent is less than the profit made from the 204 orders. Finally, email has the highest ratio, most likely because it reaches details acquired from previous customers.

4.2. Ratio

This brings us to the ratios. We table the actions in our workbook for better presentation, and through this we also calculate the view-to-purchase ratio for each traffic source.

Table 1. Cross Dimensional Analysis

Traffic Source	View	Cart	Checkout	Info	Purchase	Ratio
Email	522	326	288	190	177	0.339
Organic	2038	669	473	381	343	0.168
Paid Ads	968	358	261	216	204	0.211
Social Media	1472	200	141	112	102	0.069

This confirms our previous statements: emails convert a third of the time, whereas most people who view products through social media do not go any further.

5. SITE PERFORMANCE

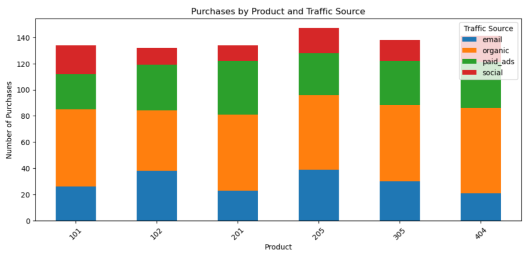
Some of the statistical calculations that have just been used can also be used on a larger scale to understand consumers on a wider scale, and how they move throughout the site.

- Items to Cart: 0.310
- Cart to Checkout: 0.710
- Checkout to Information: 0.815
- Finish Checkout: 0.919

Not only can we understand customer behaviour, but through that we can also find any problems in the site. If we look at the final ratio of 0.919 we can deduce there are no problems with the site payment process. The cart to checkout is surprisingly high as you would expect most customers would be lost leaving products in cart. The only low ratio would be adding items into cart. Explained by either customers comparing products, just looking, or the site isn't as appeasing to them.

6. PRODUCT PERFORMANCE

The site contains various different products, instead of simply counting the amount of product purchases, we can cross analyse with the traffic sources for a more detailed comparison.



All products have similar purchase amounts, with product '205' being the most popular, but the fact they are similar suggests some sort of correlation or underlying reason. This would be better understood through further data collection.

Table 2. Cross Dimensional Analysis

Traffic Source	101	102	201	205	305	404
Email	14.689266	21.468927	12.994350	22.033898	16.949153	11.864407
Organic	17.201166	13.411079	16.909621	16.618076	16.909621	18.950437
Paid Ads	13.235294	17.156863	20.098039	15.686275	16.666667	17.156863
Social Media	21.568627	12.745098	11.764706	18.627451	15.686275	19.607843

From the data we already have, we can formulate a table to see the proportion of orders spent per product by each source. This way we can see trends such as social media purchasing less of products '102' and '201' and '305' to the others.

7. CONCLUSIONS

7.1. Financial

We find that order values range between 20.18 and 299.61, with a mean order value of 106.51. If high amounts are spent on advertising and social media management to acquire a customer, then it is likely losing money on those specific transactions. That being said, large amounts of orders are from organic views and it is unlikely that the total amount spent exceeds 87975.11.

Based on the time graphs, business is consistent, and not losing clientele. However it would be ideal to see growth.

7.2. Marketing

In continuation, social media brings the most "window shoppers", so it would make sense to switch objectives and drive in a different audience, or at least cut costs. This depends on the firm's goals; social media is still a good way to build popularity.

Since emails bring in the highest ratio, working on furthering the email directives could drive a lot more sales.

7.3. Website

From baseline conversion rates, the website works well from the cart onwards. Meaning the technical payment flow is functional and not many are deterred from it.

It could be worth looking deeper into item view design, perhaps using an A/B testing method.